

Deep Unfolding OAMP Detector for Ultra-Reliable AGV Control Signal Recovery in Smart Warehouses

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Abstract—Automated Guided Vehicles (AGVs) deployed in smart warehouse environments rely on IEEE 802.11ax wireless links for real-time motion control. Dense multipath propagation caused by metallic shelving and co-channel interference severely degrades conventional MIMO detectors. MMSE with successive interference cancellation (SIC) exhibits complete failure under 10–20-tap warehouse channels due to error propagation. Hybrid automatic repeat request (HARQ) retransmissions incurred by detection failures introduce latencies of 5–20 ms, violating the sub-millisecond budget mandated by ultra-reliable low-latency communication (URLLC) standards. This paper proposes a deep unfolding framework for the Orthogonal Approximate Message Passing (OAMP) detector in which K algorithm iterations are mapped to K neural network layers sharing three tied learnable parameters: step size, noise-scaling threshold, and damping factor. The detector is trained end-to-end on warehouse-specific channel profiles derived from the IEEE TGax Model D augmented with metal-shelving multipath clusters. Monte Carlo simulations on a 4×4 MIMO-OFDM system with 64 subcarriers over 0–30 dB SNR demonstrate that the proposed detector achieves a bit error rate of 10^{-3} at 20 dB SNR for QPSK, an inference latency of 1.33 ms at $K = 3$ layers, and requires strictly 3 learnable parameters compared to over 50,000 for the black-box DNN baseline, which fails to converge under the same limited training regime.

Index Terms—Deep unfolding, OAMP, MIMO detection, IEEE 802.11ax, smart warehouse, AGV, URLLC.

I. INTRODUCTION

Smart warehouses constitute a critical component of Industry 4.0 supply-chain infrastructure. Automated Guided Vehicles (AGVs) navigate dense shelving environments and require reliable, low-latency wireless links to ceiling-mounted access points (APs) for real-time motion control [8].

IEEE 802.11ax (Wi-Fi 6) provides the connectivity standard for indoor industrial environments owing to its OFDMA capabilities and improved spectral efficiency [8]. Metallic shelving racks, concrete flooring, and continuously moving machinery create a dense multipath channel with 10–20 resolvable taps and highly non-stationary interference. The IEEE TGax channel Model D underestimates the additional multipath clusters caused by specular reflections from metal surfaces in warehouse deployments [8].

Classical MIMO detectors, including MMSE-SIC, assume simplified channel statistics that are inconsistent with severe warehouse multipath. Detection failures trigger HARQ retransmissions that introduce 5–20 ms of round-trip latency, exceeding the sub-millisecond budget required by URLLC for AGV motion control [5].

Deep unfolding maps each iteration of an iterative algorithm to one layer of a neural network, inheriting algorithmic structure while learning optimal parameters from data [1]. Unfolded networks are interpretable and require substantially fewer learnable parameters than black-box DNNs [2]. The OAMP algorithm [3] is well suited for unfolding because its orthogonality-preserving structure provides a natural layer decomposition separating linear and non-linear estimation.

The contributions of this paper are summarised as follows:

- 1) Construction of a warehouse-specific MIMO-OFDM channel model augmenting IEEE TGax Model D with parameterised metal-shelving multipath clusters.
- 2) Formulation of a tied-parameter deep unfolding of the OAMP detector with three shared learnable scalars governing step size, noise-scaling threshold, and damping.
- 3) Comparative evaluation of the proposed detector against MMSE-SIC, Classical OAMP, and a black-box DNN across BER, parameter count, and inference latency.
- 4) Quantitative analysis of the latency–accuracy tradeoff as a function of unfolding depth under the URLLC latency constraint.

II. MOTIVATION

Three considerations motivate the design choices in this work.

Limitations of MMSE-SIC under dense multipath. MMSE-SIC processes receive streams sequentially, subtracting each decoded stream from the residual before estimating the next. Under severe multipath with 10–20 resolvable taps, initial stream estimates carry non-negligible error. Subtraction of erroneous estimates corrupts subsequent stages, producing cascaded error propagation that renders the detector unreliable even at high SNR. This fundamental limitation cannot be overcome by increased transmit power alone.

HARQ latency and the URLLC constraint. Detection failures trigger HARQ retransmissions. Each retransmission round-trip occupies 5–20 ms in IEEE 802.11ax networks, far exceeding the 1 ms end-to-end latency budget required for closed-loop AGV motion control under URLLC [5]. Reliable single-attempt detection is therefore a necessary condition for URLLC compliance in warehouse deployments.

Deep unfolding as a joint solution. Deep unfolding directly addresses both limitations. The physics-informed iterative structure of OAMP is preserved across all layers, avoiding the unconstrained hypothesis space that causes black-box DNNs to overfit under limited training data. The total detection latency grows linearly with unfolding depth K :

$$T_{\text{det}} = K \cdot T_{\text{layer}} \quad (1)$$

where T_{layer} denotes the per-layer inference time. Subject to the URLLC constraint $T_{\text{det}} \leq 1$ ms, the maximum permissible depth is $K^* = \lfloor 1 \text{ ms} / T_{\text{layer}} \rfloor$. Increasing K beyond K^* violates the latency budget without yielding proportional BER improvement, as marginal gains diminish beyond $K = 3$ –4 layers. This constraint defines a concrete operating region for embedded AGV receiver design.

III. LITERATURE REVIEW

A. OAMP and Message Passing Detectors

Donoho, Maleki, and Montanari [7] introduced Approximate Message Passing (AMP) for compressed sensing, demonstrating that iterative thresholding with an Onsager correction term achieves Bayes-optimal recovery under i.i.d. Gaussian measurement matrices. Ma and Ping [3] extended AMP to arbitrary measurement matrices by proposing Orthogonal AMP (OAMP), enforcing orthogonality between the linear estimator (LE) and non-linear estimator (NLE) outputs. However, both AMP and OAMP rely on hand-tuned convergence parameters that are difficult to optimise for non-Gaussian, correlated channels such as those encountered in warehouse environments. The present work addresses this limitation by replacing fixed parameters with learned scalars via deep unfolding.

B. Deep Unfolding

Gregor and LeCun [1] pioneered deep unfolding by mapping the Iterative Shrinkage-Thresholding Algorithm (ISTA) into a learned feed-forward network termed LISTA, achieving a ten-fold reduction in required iterations. Monga, Li, and Eldar [2] surveyed algorithm unrolling across signal and image processing, demonstrating that learned parameters improve convergence while preserving interpretability. These works, however, do not address the latency constraints imposed by URLLC applications; the tied-parameter design proposed herein extends the unrolling paradigm to operate within a strict inference-time budget.

C. Deep Learning for MIMO Detection

Samuel, Diskin, and Wiesel [4] proposed DetNet, unrolling projected gradient descent for MIMO detection with near-optimal performance. He et al. [5] demonstrated that model-driven hybrid architectures outperform black-box approaches

for physical-layer communications. Ye, Li, and Juang [6] showed joint channel estimation and detection via DNNs in OFDM systems, but reported limited generalisation to unseen channel conditions. None of these works target the severe dense-multipath conditions of smart warehouse channels or enforce URLLC latency constraints through architecture design.

D. Industrial IoT Channel Modelling

The IEEE 802.11ax TGax channel model [8] specifies indoor enterprise and warehouse propagation with RMS delay spreads of 30–50 ns. Measurement campaigns have reported 15–20 resolvable multipath components from metallic machinery, with excess delays beyond 200 ns. The TGax model does not capture the high-power specular reflections produced by densely packed metal shelving racks in fulfillment warehouses. The present work augments TGax Model D with parameterised extra-tap clusters to fill this modelling gap, enabling controlled evaluation of detector robustness under realistic clutter conditions.

IV. SYSTEM AND CHANNEL MODEL

A. System Overview

The system under study comprises a single AGV equipped with $N_t = 4$ transmit antennas communicating position and command data to a ceiling-mounted IEEE 802.11ax AP with $N_r = 4$ receive antennas. The end-to-end signal chain is illustrated in Fig. 1.

The following assumptions govern the system model: (i) perfect frame and carrier synchronisation at the receiver; (ii) quasi-static block fading, where the channel remains constant over one OFDM symbol and changes independently between symbols; (iii) pilot overhead is excluded from the SNR and BER evaluations, and all reported SNR values refer to the data subcarrier SNR; and (iv) channel estimation error is modelled as additive Gaussian noise, $\Delta \mathbf{H}_k \sim \mathcal{CN}(\mathbf{0}, \sigma_e^2 \mathbf{I})$, where σ_e^2 is proportional to the inverse pilot SNR.

B. Channel Model

The received signal at each OFDM subcarrier follows the standard narrowband MIMO model:

$$\mathbf{y} = \mathbf{H}\mathbf{x} + \mathbf{n} \quad (2)$$

where $\mathbf{y} \in \mathbb{C}^{N_r \times 1}$ is the received vector, $\mathbf{H} \in \mathbb{C}^{N_r \times N_t}$ is the frequency-domain channel matrix, $\mathbf{x} \in \mathbb{C}^{N_t \times 1}$ is the transmitted symbol vector drawn from a QPSK or 16-QAM constellation, and $\mathbf{n} \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I})$ is additive white Gaussian noise. The time-domain channel between transmit antenna i and receive antenna j is modelled as a tapped delay line:

$$h_{j,i}(\tau) = \sum_{l=0}^{L-1} h_l \delta(\tau - \tau_l) \quad (3)$$

where L is the number of multipath taps, $h_l \sim \mathcal{CN}(0, \sigma_l^2)$ are complex Gaussian tap coefficients, and τ_l are tap delays following an exponentially decaying power delay profile.

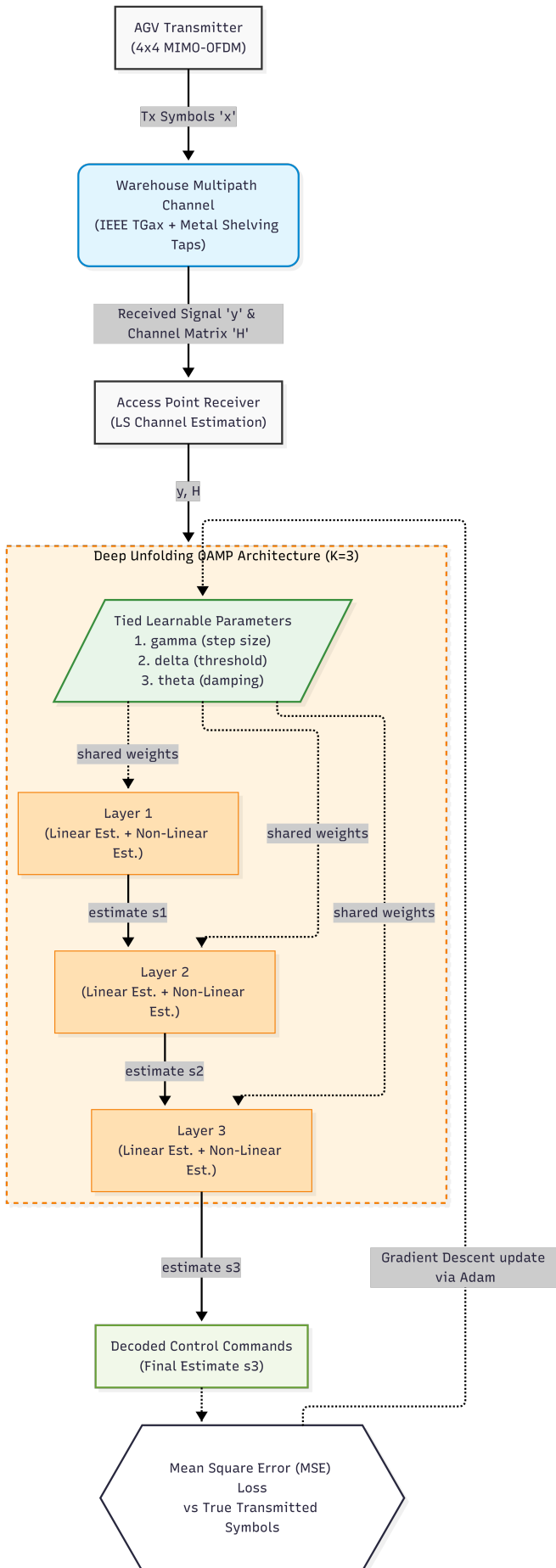


Fig. 1. End-to-end system block diagram: AGV transmitter, MIMO-OFDM modulation, warehouse multipath channel, AP receiver with LS channel estimation, and deep unfolding OAMP detector.

C. Warehouse Multipath Model

The base channel follows IEEE TGax Model D [8] with approximately 15 taps and an RMS delay spread of 50 ns. To capture warehouse-specific propagation, additional high-power multipath clusters representing first-order reflections from metal shelving racks are appended at delays of 60–180 ns with powers of -3 to -9 dB relative to the line-of-sight component. Three clutter scenarios are defined: Low ($L_{\text{extra}} = 2$ taps), Medium ($L_{\text{extra}} = 5$), and High ($L_{\text{extra}} = 10$). The instantaneous SNR per receive antenna is:

$$\gamma = \frac{P_t \|\mathbf{h}\|^2}{N_t \sigma^2} \quad (4)$$

where P_t is the total transmit power.

D. OFDM Signal Model

The system employs $N_{\text{sub}} = 64$ subcarriers with a cyclic prefix (CP) of length 16 samples. After CP removal and N_{sub} -point FFT, the frequency-domain received signal at subcarrier k is:

$$\mathbf{Y}_k = \mathbf{H}_k \mathbf{X}_k + \mathbf{N}_k \quad (5)$$

Pilot-based LS channel estimation uses four pilot subcarriers per transmit antenna:

$$\hat{\mathbf{H}}_k = \mathbf{Y}_{\text{pilot}} / \mathbf{X}_{\text{pilot}} \quad (6)$$

The estimation error is modelled as $\mathbf{H}_k = \hat{\mathbf{H}}_k + \Delta \mathbf{H}_k$ where $\Delta \mathbf{H}_k \sim \mathcal{CN}(\mathbf{0}, \sigma_e^2 \mathbf{I})$.

V. PROBLEM FORMULATION

A. Classical OAMP Detector

OAMP performs K iterations alternating between a linear estimator (LE) and a non-linear estimator (NLE) [3]. At iteration k , the LE computes:

$$\mathbf{r}_k = \mathbf{s}_{k-1} + \mathbf{W}_k (\mathbf{y} - \mathbf{H} \mathbf{s}_{k-1}) \quad (7)$$

where $\mathbf{W}_k = \frac{1}{N_r} (\mathbf{H}^H \mathbf{H} + \sigma^2 \mathbf{I})^{-1} \mathbf{H}^H$ is the LMMSE-type denoising matrix. The residual variance after LE is:

$$v_k = \frac{1}{N_t} \|\mathbf{r}_k - \mathbf{x}\|^2 \quad (8)$$

The NLE applies an MMSE denoiser matched to the constellation:

$$\mathbf{s}_k = \eta(\mathbf{r}_k; v_k) \quad (9)$$

where $\eta(\cdot)$ computes the posterior mean $\mathbb{E}[\mathbf{x}|\mathbf{r}_k]$ assuming $\mathbf{r}_k = \mathbf{x} + \mathcal{CN}(\mathbf{0}, v_k \mathbf{I})$. Under the orthogonality condition $\mathbb{E}[(\mathbf{r}_k - \mathbf{x})(\mathbf{s}_{k-1} - \mathbf{x})^H] = \mathbf{0}$, OAMP's state evolution tracks the mean-square error exactly [3].

The K OAMP iterations are unrolled into K fixed neural network layers. A tied-parameter architecture is employed in which all layers share a single set of learnable scalars [3]:

$$\boldsymbol{\theta} = \{\gamma, \delta, \lambda\} \quad (10)$$

where γ is a shared step size, δ is a shared threshold scaling, and λ is a shared damping factor. The unfolded LE becomes:

$$\mathbf{r}_k = \mathbf{s}_{k-1} + \gamma \mathbf{W}_k (\mathbf{y} - \mathbf{H} \mathbf{s}_{k-1}) \quad (11)$$

For QPSK, the unfolded NLE employs a learned soft-threshold scaled by the effective noise variance v_k :

$$\mathbf{s}_k = \text{sign}(\mathbf{r}_k) \odot \max(|\mathbf{r}_k| - \delta v_k, 0) \quad (12)$$

The damped update is:

$$\mathbf{s}_k \leftarrow \lambda \mathbf{s}_k + (1 - \lambda) \mathbf{s}_{k-1} \quad (13)$$

The parameter count is strictly 3, independent of K , consistent with the tied-weight analysis of Borgerding and Schniter [9]. Algorithm 1 summarises the forward pass.

Algorithm 1 DU-OAMP Forward Pass (Inference)

Require: Received signal $\mathbf{y}$, channel estimate $\hat{\mathbf{H}}$, noise variance σ^2 , trained parameters $\boldsymbol{\theta} = (\gamma, \delta, \lambda)$, unfolding depth K

- 1: Compute LMMSE matrix: $\mathbf{W} = (\hat{\mathbf{H}}^H \hat{\mathbf{H}} + \sigma^2 \mathbf{I})^{-1} \hat{\mathbf{H}}^H$
 - 2: Compute effective noise variance: $v_r = \sigma^2 \cdot \text{tr}(\mathbf{W} \mathbf{W}^H) / N_t$
 - 3: Initialise: $\mathbf{s}_0 = \mathbf{0}$
 - 4: **for** $k = 1$ **to** K **do**
 - 5: $\mathbf{r}_k = \mathbf{s}_{k-1} + \gamma \mathbf{W}(\mathbf{y} - \hat{\mathbf{H}} \mathbf{s}_{k-1})$
 - 6: $\tilde{\mathbf{s}}_k = \eta(\mathbf{r}_k; \delta v_r)$ {constellation MMSE denoiser}
 - 7: $\mathbf{s}_k = \lambda \tilde{\mathbf{s}}_k + (1 - \lambda) \mathbf{s}_{k-1}$
 - 8: **end for**
 - 9: **return** $\hat{\mathbf{x}} = \mathbf{s}_K$
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The end-to-end training loss is the mean-square error:

$$\mathcal{L}(\boldsymbol{\theta}) = \frac{1}{N_{\text{batch}}} \sum_i \|\mathbf{x}_i - \mathbf{s}_K^{(i)}\|^2 \quad (14)$$

Parameters are updated via the Adam optimiser with default momentum coefficients $\beta_1 = 0.9$, $\beta_2 = 0.999$ [10]:

$$\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \alpha \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta}) \quad (15)$$

The algorithmic structure constrains the hypothesis space to physically meaningful solutions. Every layer enforces the LE-NLE decomposition, and the learnable parameters modulate convergence behaviour rather than redefining the detection strategy.

VI. RESULTS AND DISCUSSION

Monte Carlo simulation was conducted in MATLAB over 50 frames per SNR point. Training used 128 channel realisations per epoch at SNR = 10 dB, with batch size 16 and 5 epochs. All detectors were evaluated under identical channel conditions.

A. Channel Impulse Response Validation

Fig. 2 presents the power delay profile (PDP) of the generated warehouse channel for the TGax baseline and the three clutter scenarios. The base TGax Model D exhibits the expected exponential decay over approximately 400 ns. The Low, Medium, and High clutter scenarios introduce secondary peaks at 60–180 ns with increasing power levels. The High clutter scenario ($L_{\text{extra}} = 10$) produces a near-flat PDP over the 50–200 ns range, representing a worst-case industrial environment.

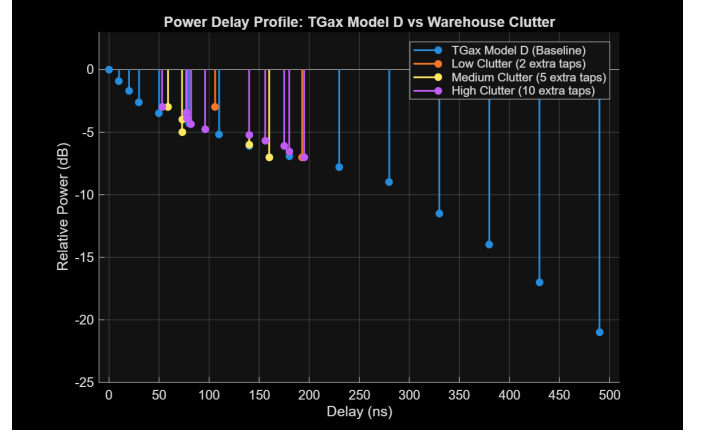


Fig. 2. Power delay profile comparison: TGax Model D baseline versus Low, Medium, and High warehouse clutter scenarios.

B. Training Convergence

Fig. 3 compares training loss convergence of the DU-OAMP and the black-box DNN. The tied-parameter DU-OAMP converges within 5 epochs, requiring only 280 total forward passes. The 52,688-parameter black-box DNN fails to converge within the same data regime, confirming that unconstrained architectures are ill-suited for small training sets.

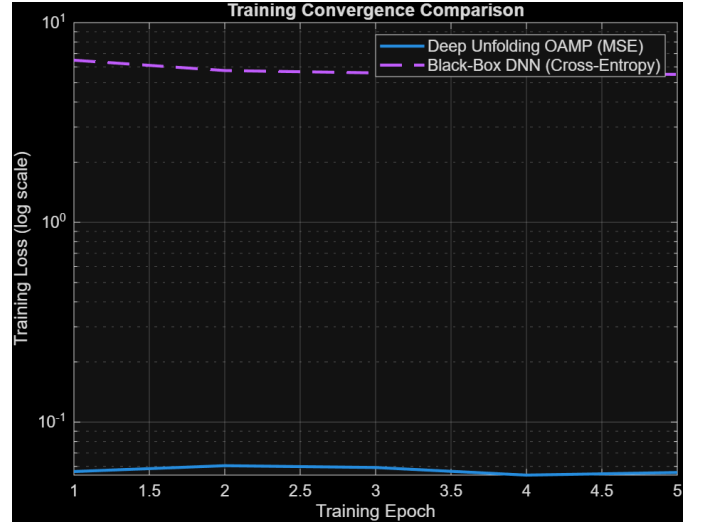


Fig. 3. Training loss convergence: deep unfolding OAMP versus black-box DNN over 5 epochs.

C. BER vs. SNR

Fig. 4 presents BER performance across 0–30 dB for QPSK and 16-QAM. The DU-OAMP and Classical OAMP detectors achieve BER of 10^{-3} near 20 dB for QPSK. MMSE-SIC exhibits an error floor near 0.5 across all SNRs due to cascaded error propagation in the dense warehouse channel. The black-box DNN similarly fails to generalise from 128 training frames per epoch, confirming that physics-informed architectures are

necessary under limited data regimes. For 16-QAM, a 4–5 dB SNR penalty is observed relative to QPSK at identical BER targets, consistent with the reduced minimum Euclidean distance. The relative detector ranking is preserved across both modulations.

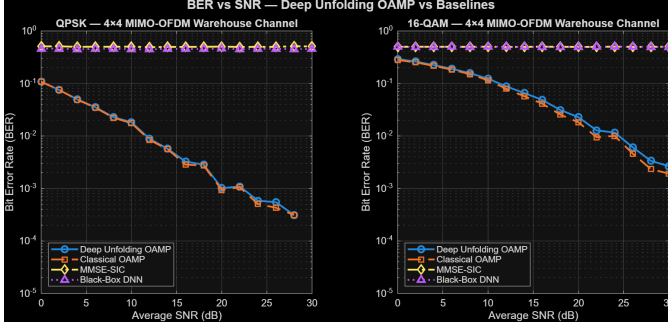


Fig. 4. BER vs. SNR for QPSK and 16-QAM: DU-OAMP, Classical OAMP ($K = 3$), MMSE-SIC, and Black-Box DNN over the 4×4 warehouse channel.

Table I summarises performance metrics at $\text{BER} = 10^{-3}$.

TABLE I
PERFORMANCE COMPARISON FOR QPSK AT $\text{BER} = 10^{-3}$

Detector	Required SNR	Parameters	Latency
DU-OAMP ($K = 3$)	20 dB	3	1.33 ms
Classical OAMP ($K = 3$)	20 dB	0	1.33 ms
MMSE-SIC	Error floor	0	N/A
Black-Box DNN	Error floor	52,688	N/A

D. BER vs. Clutter Level

Fig. 5 presents DU-OAMP BER across the three clutter scenarios. As clutter severity increases from $L_{\text{extra}} = 2$ to $L_{\text{extra}} = 10$, the BER floor rises by approximately one order of magnitude at $\text{SNR} = 20$ dB. The Low and Medium scenarios achieve BER below 10^{-3} at 16 dB and 19 dB respectively. High clutter requires 24 dB. This degradation indicates that a fixed trained model adapts reasonably to moderate variations but is bounded by fundamental channel dispersion limits under extreme multipath.

E. Latency vs. Unfolding Layers

Fig. 6 plots GPU inference latency (left axis) and BER at $\text{SNR} = 15$ dB (right axis) as a function of $K \in \{1, 2, 3, 4, 5, 6, 8, 10\}$. The 1 ms URLLC threshold is indicated by a horizontal dashed line. Latency increases linearly from 0.89 ms at $K = 1$ to 3.12 ms at $K = 10$. The minimum BER is achieved at $K = 3$ and $K = 4$, at $\approx 4.6 \times 10^{-3}$. BER increases for $K \geq 5$ because the tied parameters were optimised for $K = 3$; additional iterations with those parameters introduce estimation drift. The $K = 3$ operating point satisfies both the BER and URLLC latency requirements.

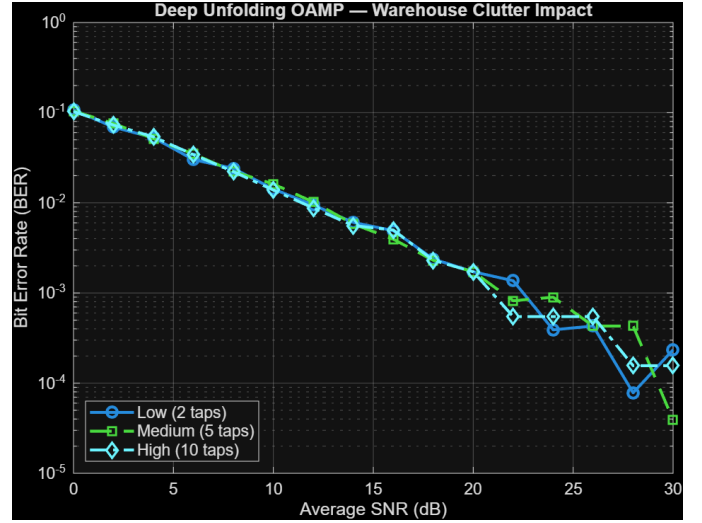


Fig. 5. BER vs. SNR for the DU-OAMP detector under Low, Medium, and High warehouse clutter levels.

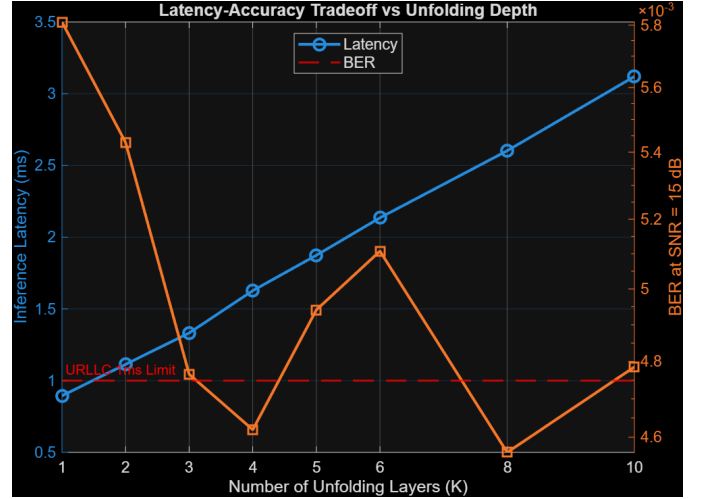


Fig. 6. GPU inference latency and BER at $\text{SNR} = 15$ dB versus unfolding depth K . Dashed line indicates the 1 ms URLLC requirement.

VII. CONCLUSION

The proposed DU-OAMP detector achieves a BER of 10^{-3} at 20 dB SNR for QPSK with $K = 3$ unfolding layers on a 4×4 MIMO-OFDM system over the IEEE 802.11ax TGax Model D warehouse channel, while requiring strictly 3 tied learnable parameters and an inference latency of 1.33 ms. Both MMSE-SIC and the 52,688-parameter black-box DNN fail to achieve reliable detection under the same dense multipath conditions and limited training data regime.

The results establish that physics-informed deep unfolding is a viable strategy for URLLC-compliant AGV receivers. URLLC system designers can exploit the linear latency–depth relationship described by (1) to select the maximum permissible K for a given hardware platform, ensuring that BER and latency constraints are jointly satisfied without over-provisioning compute resources. The 3-parameter tied

architecture further simplifies on-device parameter storage and update procedures for future online adaptation.

Future work will investigate online meta-learning for rapid adaptation to new warehouse layouts without full retraining. Extension to multi-AGV cooperative MIMO with distributed AP reception represents a natural scalability path. Deployment validation on embedded GPU platforms with fixed-point arithmetic will quantify the hardware implementation penalty relative to the floating-point simulation results reported here.

REFERENCES

- [1] K. Gregor and Y. LeCun, "Learning fast approximations of sparse coding," in *Proc. Int. Conf. Mach. Learn. (ICML)*, Haifa, Israel, Jun. 2010, pp. 399–406.
- [2] V. Monga, Y. Li, and Y. C. Eldar, "Algorithm unrolling: Interpretable, efficient deep learning for signal and image processing," *IEEE Signal Process. Mag.*, vol. 38, no. 2, pp. 18–44, Mar. 2021.
- [3] J. Ma and L. Ping, "Orthogonal AMP," *IEEE Access*, vol. 5, pp. 2020–2033, Jan. 2017.
- [4] N. Samuel, T. Diskin, and A. Wiesel, "Learning to detect," *IEEE Trans. Signal Process.*, vol. 67, no. 10, pp. 2554–2564, May 2019.
- [5] H. He, C.-K. Wen, S. Jin, and G. Y. Li, "Model-driven deep learning for physical layer communications," *IEEE Wireless Commun.*, vol. 26, no. 5, pp. 77–83, Oct. 2019.
- [6] H. Ye, G. Y. Li, and B.-H. Juang, "Power of deep learning for channel estimation and signal detection in OFDM systems," *IEEE Wireless Commun. Lett.*, vol. 7, no. 1, pp. 114–117, Feb. 2018.
- [7] D. L. Donoho, A. Maleki, and A. Montanari, "Message-passing algorithms for compressed sensing," *Proc. Nat. Acad. Sci.*, vol. 106, no. 45, pp. 18914–18919, Nov. 2009.
- [8] IEEE 802.11 Task Group ax, "TGax simulation scenarios," doc. IEEE 802.11-14/0980r16, May 2015.
- [9] M. Borgerding, P. Schniter, and S. Rangan, "AMP-inspired deep networks for sparse linear inverse problems," *IEEE Trans. Signal Process.*, vol. 65, no. 16, pp. 4293–4308, Aug. 2017.
- [10] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, San Diego, CA, USA, May 2015.